



GEOOptimize Inc
1-204-318-2156

220 Portage Avenue, Suite 1410
Winnipeg, Manitoba
Canada, R3C 0A5

Proposal for:
Aaron Schauger
Energy Engineer
Labella Associates
300 Pearl St Suite 130
Buffalo, NY
14202

Proposal Date:
2023-02-08

Proposal Number:
QU-070

Reference
US NY Labella – Central Hudson

Understanding of Project

In November of 2021, the New York State Public Service Commission (“Commission”) approved the Company’s 2021-2024 Rate Plan. Within the rate plan, the Commission directed Central Hudson to conduct a Geothermal District Loop Study (“Geothermal Study”) with the purpose of identifying areas of the Company’s service territory where the potential for geothermal district loop (“GDL”) pilots might exist.

Central Hudson is aiming to contribute to aforementioned policy goals by identifying opportunities for GDL projects. Central Hudson has requested a study that identifies potential opportunities throughout the entire Central Hudson Gas and Electric service territory and where practical give additional focus to areas within Disadvantaged Communities (“DAC”). The purpose of the project is to conduct a cost-effective approach to identify potential GDL pilot project locations within the Central Hudson Gas and Electric service territory. Labella has retained GEOptimize to assist in the detailed district energy modelling and district ground source heat pump (GSHP) system design of one site. Labella has also requested training and knowledge transfer on GEOptimize’s district GSHP system design methodology.

Proposed Design Methodology

District Geothermal System Modelling

To model and design the layout of the GSHP district energy system, TRNSYS V18.03 will be used. TRNSYS is a simulation program used to model large, interconnected energy systems. Energy loads for each individual building provided by Labella will be imported into the district energy model to simulate the building loads and their impact on the district energy system. To obtain a starting point on the number of boreholes needed for the aggregate district load, GLD software will be used. Once a rough number of boreholes is determined, TRNSYS software is then used to analyze the relationship

between the GHX's, buildings, and other potential energy resources. Other energy sources and / or heat sinks that can be modelled with TRNSYS are wastewater heat transfer, heat dissipation with a snow melt system, heat injection from solar thermal collectors, open water well heat exchangers, pond heat exchangers, heat dissipation from fluid coolers or cooling towers, etc.

A series of simulations of the model are developed to optimize system performance including:

- The size and configuration of GHX modules,
- Location of GHX modules in relation to building heating and cooling requirements,
- System control strategy,
- Additional waste and renewable energy resources and heat sinks and their impact on the size and performance of the GHX modules,

Scope of work

District Feasibility Assessment

- 1) Develop a conceptual district ambient loop GSHP system design for 1 pilot site
- 2) Develop one TRNSYS model for the pilot site
- 3) Develop a one-line system flow diagram for the pilot site
- 4) Work with Labella to determine cost estimates for the project
- 5) Work with Labella to determine energy cost and CO₂ savings based on outputs from the building energy models and TRNSYS model
- 6) Work with Labella to determine a phasing plan of the geothermal system over the course of several years, based on urgency of building conversion and location of buildings relative to underground heat source/sinks
- 7) GEOptimize has allowed for three 1-hour calls to explain TRNSYS modelling and district GSHP design methodology

Deliverables

GEOptimize will prepare a PowerPoint presentation to be delivered to Labella at a meeting via screen sharing software detailing the conceptual system design. GEOptimize will also schedule three 1-hour calls to explain TRNSYS modelling and district GSHP design methodology.

Pricing

Description	Rate	Qty	Line Total
District Feasibility Study	\$27,425.00	1	\$27,425.00
Subtotal			\$27,425.00
Proposal Total (USD)			\$27,425.00

Terms

- Additional work out of scope will be charged at \$250/hr.
- Invoices will be submitted monthly for work completed to date. Payment is expected on receipt of invoice.

Authorized signature – Labella Associates